STA 35C Statistical Data Science III

Practice midterm exam 2 – Set A

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Name: _____ **Student ID:** _____

Instructions: This midterm exam is **closed-book**, except for the permitted note sheet described below. You may bring a pen or pencil, one letter-sized sheet of *hand-written* notes (both sides), and a *non-graphing* calculator. No other materials (e.g., textbooks) are allowed. For a timed attempt, allow 50 minutes. Problems 1–4 total **120 points**; Problem 5 offers *up to 10 bonus points*. Once you receive this exam, please confirm that you have all 7 pages.

- Make sure to clearly write your name and student ID above.
- For Problem 1, give only the requested answers; no explanations are required. For Problems 2–5, show the relevant steps for partial credit. Sentence limits apply to explanations, not displayed calculations.
- If necessary, round numerical answers to three decimal places; fractions, logarithms, and exponentials may be left unevaluated.
- Complete Problems 1–4 before attempting the optional bonus questions in Problem 5.

Problem	**Score**
Problem 1	
Problem 2	
Problem 3	
Problem 4	
Problem 5 (bonus)	
Total	

Problem 1 (30 points). Basic concepts

Give only the requested answers. No explanations are required.

(a) **(5 points)** Which component of the usual squared prediction error decomposition measures **sensitivity of the fitted prediction to the training sample**? Circle one: **squared bias / variance of the fitted prediction / irreducible noise**.

(b) **(5 points)** Three fixed tuning values are evaluated by five-fold cross-validation, using direct refitting for each tuning value and fold. How many model fits are required, **excluding the final fit on all data**?

(c) **(5 points)** An ordinary bootstrap resample draws n times independently with replacement from n observed rows. Give the probability that **one specified row is never drawn**. You may leave a power unevaluated.

(d) **(5 points)** Forward stepwise selection starts with six candidate predictors and has already included two of them. How many **candidate additions** are compared at the next step?

(e) **(5 points)** Ridge regression is fitted on a fixed dataset with response mean $\overline{y}$. Its intercept is unpenalized. As $\lambda \to \infty$, what **constant fitted prediction** does it approach?

(f) **(5 points)** All 10 null hypotheses are true. Their tests are independent, and each rejects with probability **exactly** 0.05. Give the probability of **at least one false rejection**. Leave your answer as an expression if preferred.

Problem 2 (30 points). Prediction error and resampling

Each part concerns a separate analysis.

(a) (10 points) At a fixed input x_0, $f(x_0) = 3$. Across training samples, $\widehat{f}(x_0)$ takes the values $1, 2, 3$, each with probability $1/3$; these are its **only possible values**. A new response is $Y_0 = 3 + \varepsilon_0$, where ε_0 is independent of the training sample, with mean 0 and variance 1.

Compute $\mathbb{E}[(Y_0 - \widehat{f}(x_0))^2]$, showing the mean and variance of $\widehat{f}(x_0)$ in your calculation.

(b) (10 points) Ten observations are partitioned into three validation folds. Each model fit uses only the observations outside its validation fold. The held-out results are:

Validation fold	1	2	3
Number of held-out observations	2	3	5
Mean squared prediction error within the fold	1	2	5

Compute the CV MSE per observation, pooling all held-out predictions.

(c) (10 points) An i.i.d. sample of delivery times, in minutes, is $4, 6, 8, 9, 12$. The statistic of interest is the sample median. Five bootstrap resamples, each obtained by five independent draws with replacement from these observations, are shown below in sorted order.

Resample	Sorted observations
1	$4, 4, 4, 6, 12$
2	$4, 6, 6, 8, 12$
3	$4, 6, 8, 9, 12$
4	$4, 8, 8, 9, 9$
5	$6, 8, 9, 9, 12$

Report the bootstrap standard error of the sample median and the resulting normal-approximation 95% confidence interval for the population median. Use the sample standard deviation of the five bootstrap medians (divisor $5 - 1$) and the interval rule

$$\widehat{m} \ \pm \ 1.96\, \widehat{\mathrm{SE}}_{\mathrm{boot}},$$

where $\widehat{m}$ is the **original sample median**. These five resamples illustrate the calculation only.

Problem 3 (30 points). Subset search and regularization

For parts (a)–(b), the following RSS values come from least-squares fits on the **same training data**. Every model includes an intercept; $\varnothing$ denotes no predictors.

Predictors	RSS	Predictors	RSS
$\varnothing$	240	X_1, X_2	65
X_1	72	X_1, X_3	70
X_2	135	X_2, X_3	30
X_3	198	X_1, X_2, X_3	30

(a) (10 points) For each subset size k, **write the predictor subset with the smallest RSS** in the table below. Use $\varnothing$ for $k = 0$.

Number of predictors k	Best subset at this size
0	
1	
2	
3	

(b) (10 points) Starting with the intercept-only model, apply **forward stepwise selection using training RSS** until exactly two predictors are included. Report:

(i) the selected pair, showing which predictor is added at each step;
(ii) a **one-sentence explanation** of why it differs from the best two-predictor subset in part (a).

(c) (10 points) In a **separate analysis**, lasso with a positive penalty on the slopes gives

$$\widehat{y}_{\text{lasso}} = 5 + 2x_1 + 0x_2 - x_3.$$

An analyst drops X_2 and refits a regression using only X_1 and X_3 by ordinary least squares, with an intercept and the **same training observations**.

For each statement, write **True** or **False** and give a **one-sentence justification**.

(i) The OLS refit cannot have a larger training RSS than the original lasso fit.
(ii) The OLS refit cannot have a larger expected squared prediction error on independent new observations from the same population.

Problem 4 (30 points). Reviewing a bike-rental forecast

A team predicts daily bike-rental demand. Its numerical analyses and a proposed workflow are described below.

(a) (10 points) Three ridge penalty values are evaluated using the **same CV folds**. The equations below are the corresponding fits on all development data; the intercept is unpenalized.

Penalty λ	CV MSE	Fitted equation
0	6.4	$\widehat{y} = 10 + 4x$
2	5.1	$\widehat{y} = 12 + 2x$
6	5.8	$\widehat{y} = 13 + x$

Report both quantities:

(i) the λ with the smallest CV MSE;
(ii) that setting's fitted prediction at $x = 2$.

(b) (10 points) A separate study produces the following valid p-values for six prespecified tests. These are the **entire testing family**, already in increasing order.

Null hypothesis	$H_{0,1}$	$H_{0,2}$	$H_{0,3}$	$H_{0,4}$	$H_{0,5}$	$H_{0,6}$
p-value	0.004	0.009	0.012	0.020	0.024	0.040

At family-wise level 0.05, report the rejection set for each method, showing its cutoff comparisons:

(i) **Bonferroni** (5 points);
(ii) **Holm** (5 points).

(c) (10 points) For an alternative prediction procedure, an analyst selects the two predictors with the largest absolute sample correlations with the response using **all development observations**. The analyst then holds these two predictors fixed during five-fold CV and claims that this assesses the complete selection-and-fitting procedure.

Describe the corrected procedure within each CV fold in **at most two sentences**, specifying which observations are used for selection, fitting, and evaluation.

Problem 5 (10 bonus points). Optional bonus questions

These questions are independent of the required questions and of each other.

(a) (5 points) An ordinary bootstrap resample draws n times independently with replacement from n observed rows. Rows are treated as distinct by their **indices**, even if their values agree. Let D be the number of distinct row indices represented in the resample.

Find $\mathbb{E}[D]$ conditional on the original dataset and justify your answer. You may leave a power unevaluated.

(b) (5 points) Two predictors are identical on every training observation: $x_{i1} = x_{i2}$ for all i. With $\lambda > 0$, ridge regression minimizes

$$\sum_{i=1}^{n}(y_i - b_0 - b_1 x_{i1} - b_2 x_{i2})^2 + \lambda(b_1^2 + b_2^2).$$

Must every minimizer satisfy $b_1 = b_2$? Justify your answer with a brief algebraic argument.

(Hint: Change b_1 and b_2 while keeping their sum fixed.)